



Helicopter Rotor Dynamics: Aerodynamic and Multibody Simulation in MSC ADAMS

BTP Poster

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Abstract

Helicopter rotors enable remarkably versatile flight behaviors such as hovering and vertical takeoff but with significant trade-offs- including unsteady aerodynamics and blade dynamics as well as complexity in the swashplate-rotor design and its control challenges. This thesis investigates blade flapping under aerodynamic loads for a mechanically hinged rotor swashplate model. A CAD-based rotor and swashplate system was developed and simulated in MSC ADAMS, applying both constant and time-varying angle of attack (AoA) inputs. Simulation results were validated with MATLAB-based ODE solutions, establishing a reliable multibody model for further study of rotor dynamics, including cyclic and collective pitch control.

1. Problem definition

Helicopter rotor systems rely on dynamic flapping behavior of blades to maintain lift and control. Understanding this behavior under various aerodynamic conditions is critical for accurate modeling and control design. The main objectives here are:

- To simulate flapping motion in helicopter blades with both constant and harmonically varying AoA using MSC ADAMS.
- To validate the simulation using MATLAB-based ODE solutions for flap angle behavior.
- To model a realistic rotor–swashplate system with collective and cyclic pitch potential inputs.

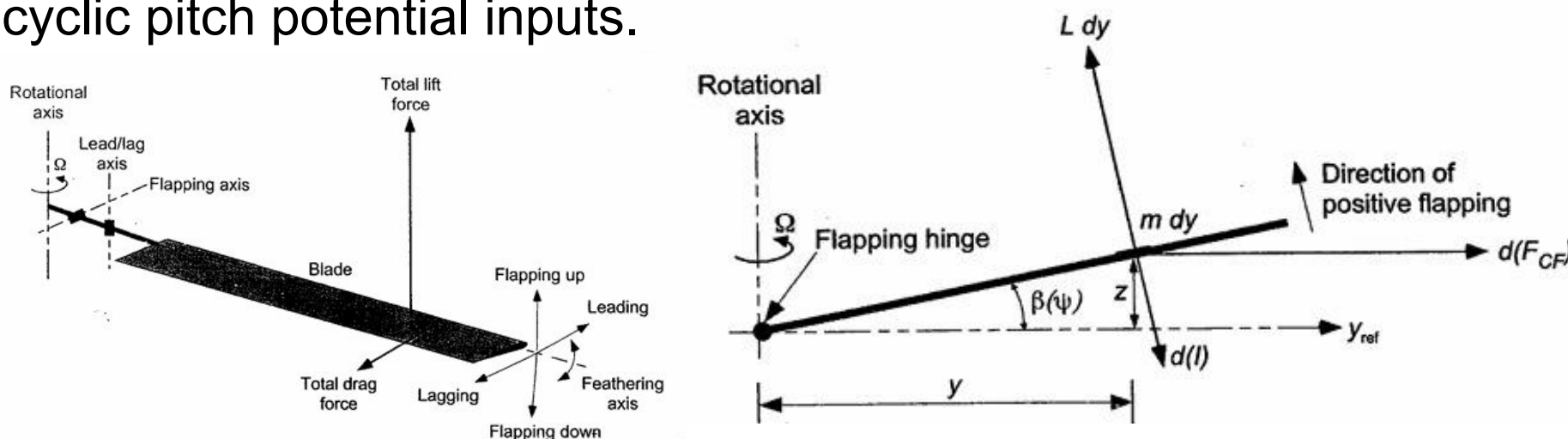


Figure 1: Blade DoFs Figure 2: Elemental Blade Flap Analysis

2. Methodology (Flow chart, etc)

- Theoretical Modeling
 - Derived ODEs for flapping blade with lift and inertial terms.
 - Assumed harmonic excitation for input time-varying AoA.
- CAD Development
 - Designed simplified single-blade and full rotor–swashplate assemblies in Autodesk Fusion 360.
- Simulation in MSC ADAMS
 - Implemented both point lift and torque-based lift (with constant and harmonic AoA) to simulate aerodynamic loads on models.
- Validation with MATLAB
 - Simulated the corresponding ODE models in MATLAB.
 - Compared flap angle, velocity over time, as well as FFTs to ensure frequency matching.

$$I_x \ddot{\beta} + (I_x + m * d^2) \Omega^2 \beta = M_\beta.$$

$$M_\beta = K * (\theta + \theta_c \cos(w_c t) - \dot{\beta} / \Omega).$$

Parameter	Value	Units
Ω (Motor speed)	25	rad/s
L (Lift)	50	N
m (Mass)	0.287286	Kg
R (Length of blade)	0.377	m
d (Eccentricity of rotating axis with flap hinge)	0.1355	m
I_x (Inertia about flap hinge x)	10	Nm^2
I_z (Inertia about flap hinge z)	10	Nm^2

Figure 3: ODEs and Params for MATLAB

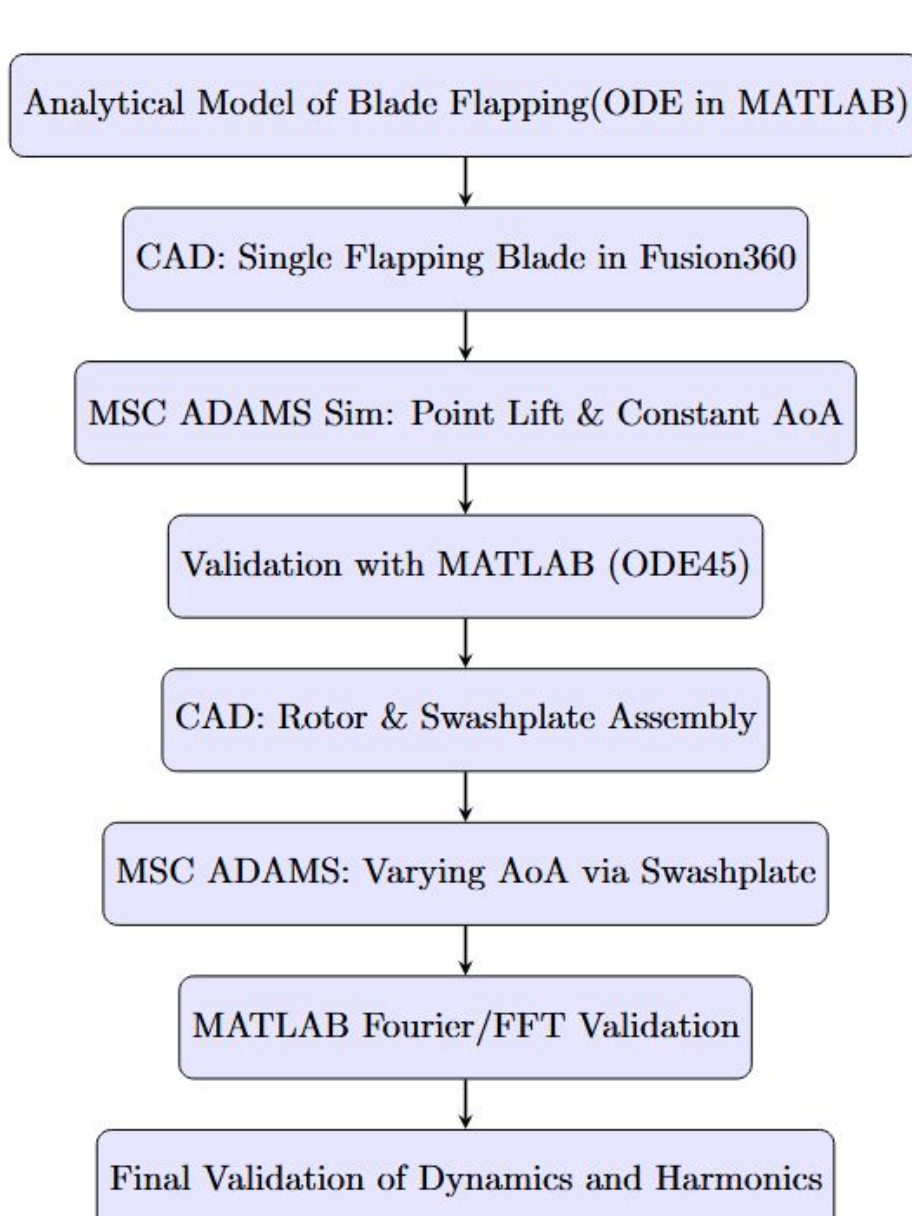


Figure 5: Methodology Flowchart

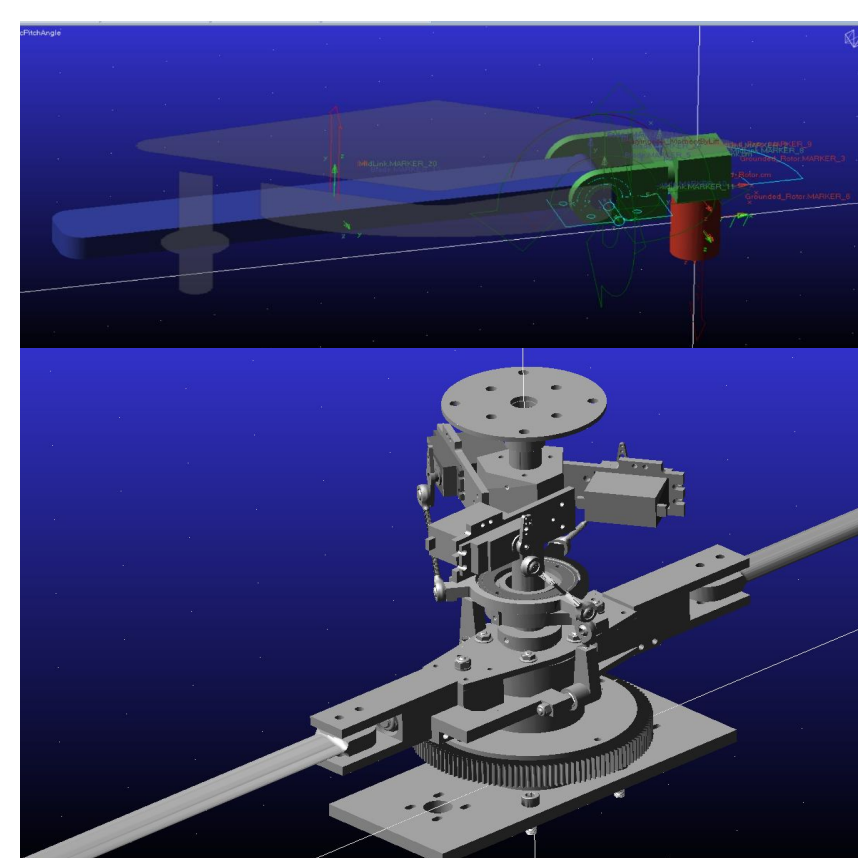


Figure 4: MSC ADAMS Models

3. Results & Discussion

- Flap Angle and Flap Velocity matched closely between ADAMS and MATLAB simulations for all 3 cases of aerodynamic loads. (Max 0.001 rad error).
- Steady-State Flap Angle Average with constant and varying AoA. matches predicted: $\beta = K * \theta / (I_x + m * d^2)$
- Harmonic AoA input showed accurate frequency response with correct amplitude (≈ 0.021 rad) and frequency (≈ 6 rad/s) via FFT (Fig 7).

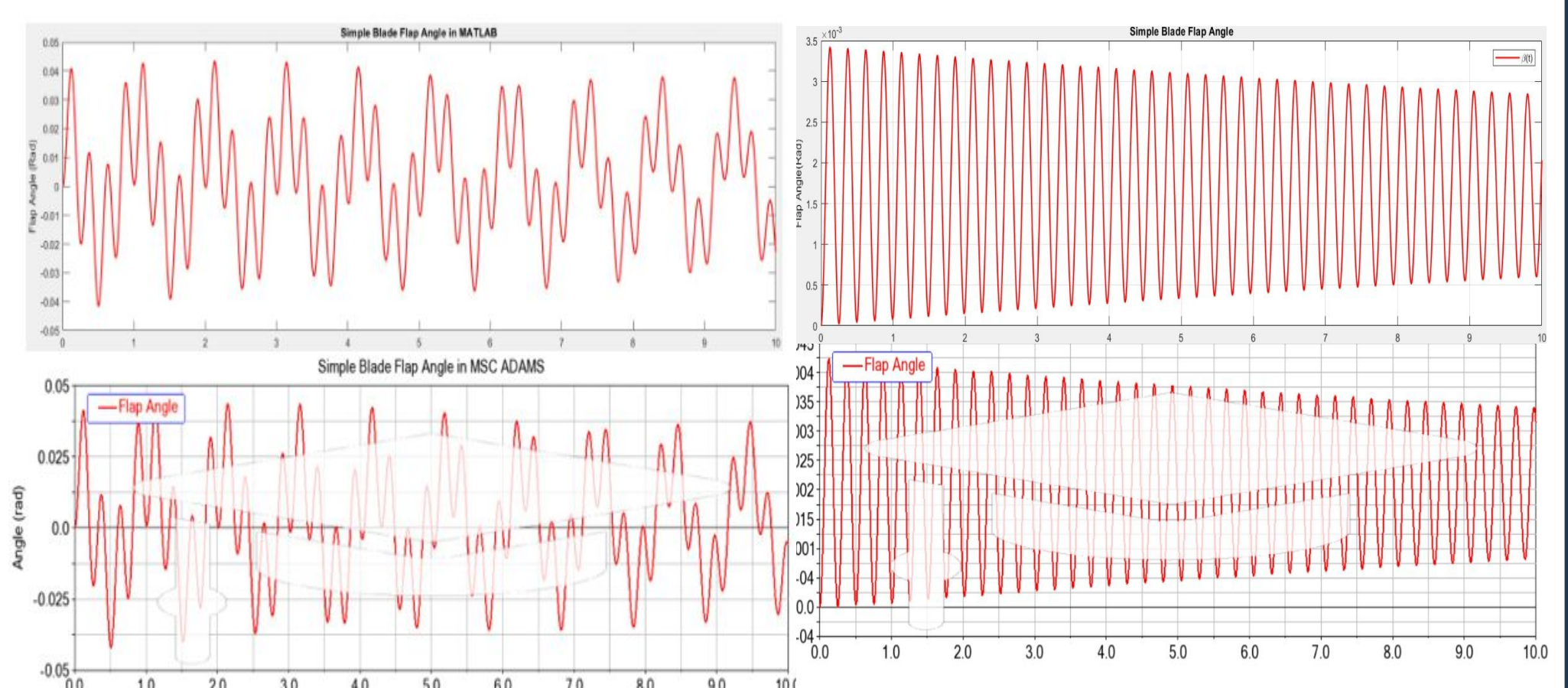


Figure 6: Flap for Harmonic AoA and Constant AoA- MATLAB vs ADAMS

- MSC ADAMS captured both transients and steady-state behavior accurately: Amplitudes, Frequencies, Harmonic Coefficients (See Fig 6)
- The rotor–swashplate system's motion confirmed theoretical expectations and is now a strong base for future cyclic/collective studies.

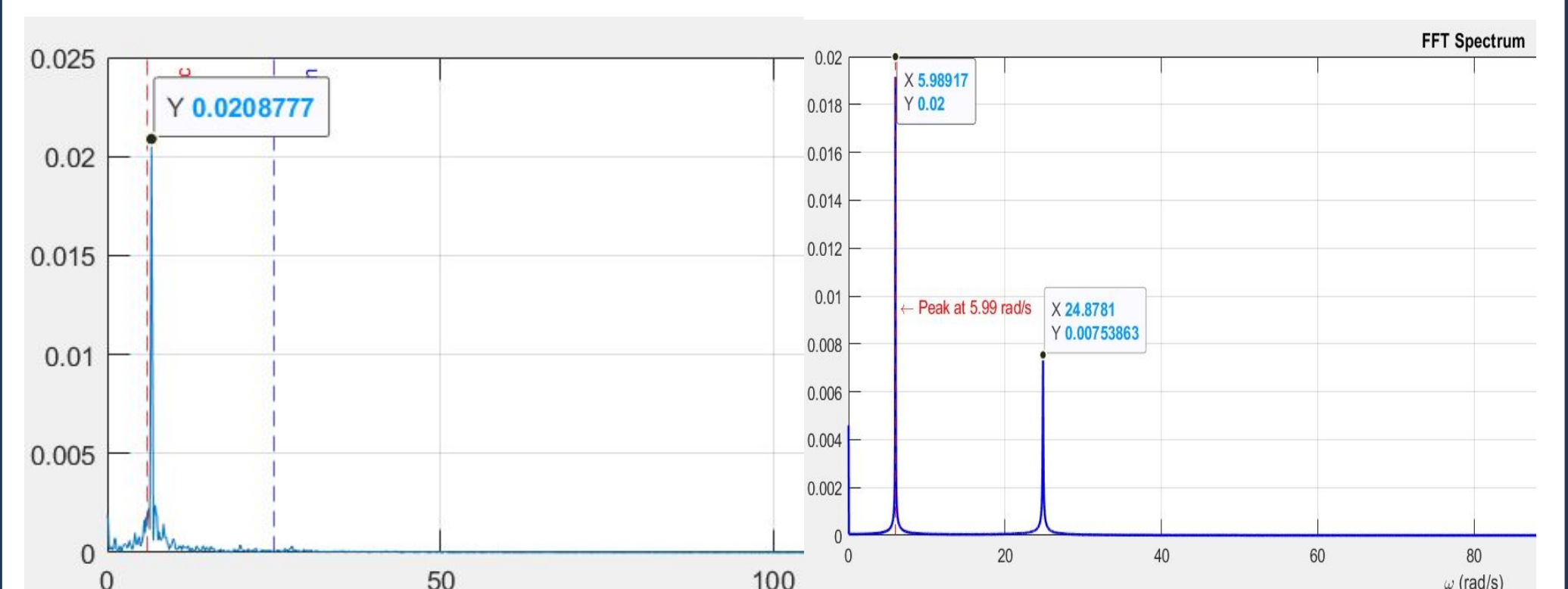


Figure 7: FFTs for Harmonic AoA for Analytical and Simulation Solutions

4. Conclusions

- This study successfully models and validates helicopter blade flapping dynamics under different AoA profiles through a combination of ODE-based formulation and ADAMS simulations showing excellent model fidelity.
- The rotor–swashplate CAD setup is validated for collective pitch behavior and ready for extension into cyclic pitch control and flexible blade analysis.
- Future work may explore advanced aerodynamics, forward flight regimes, and control algorithm design.

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